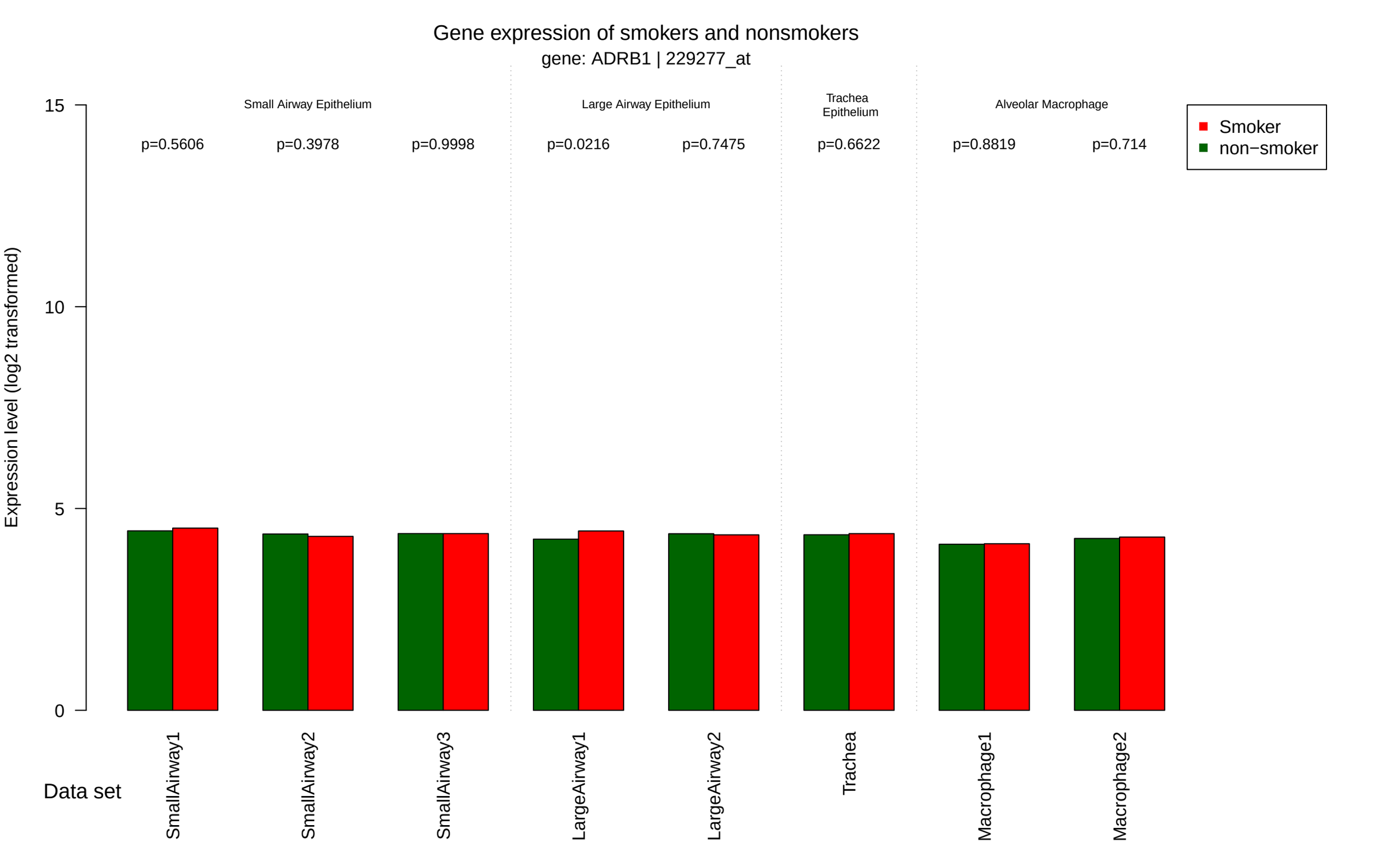
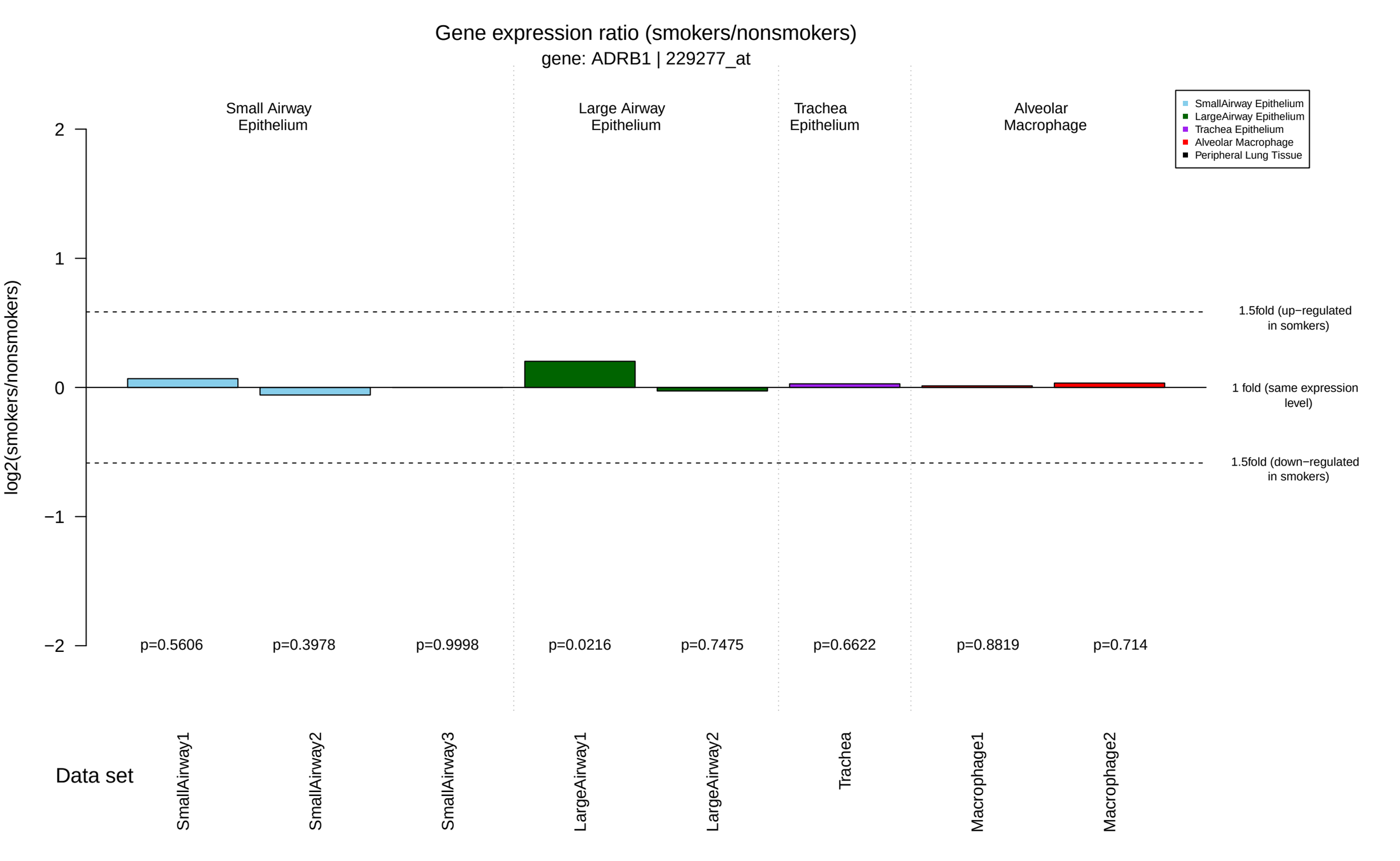
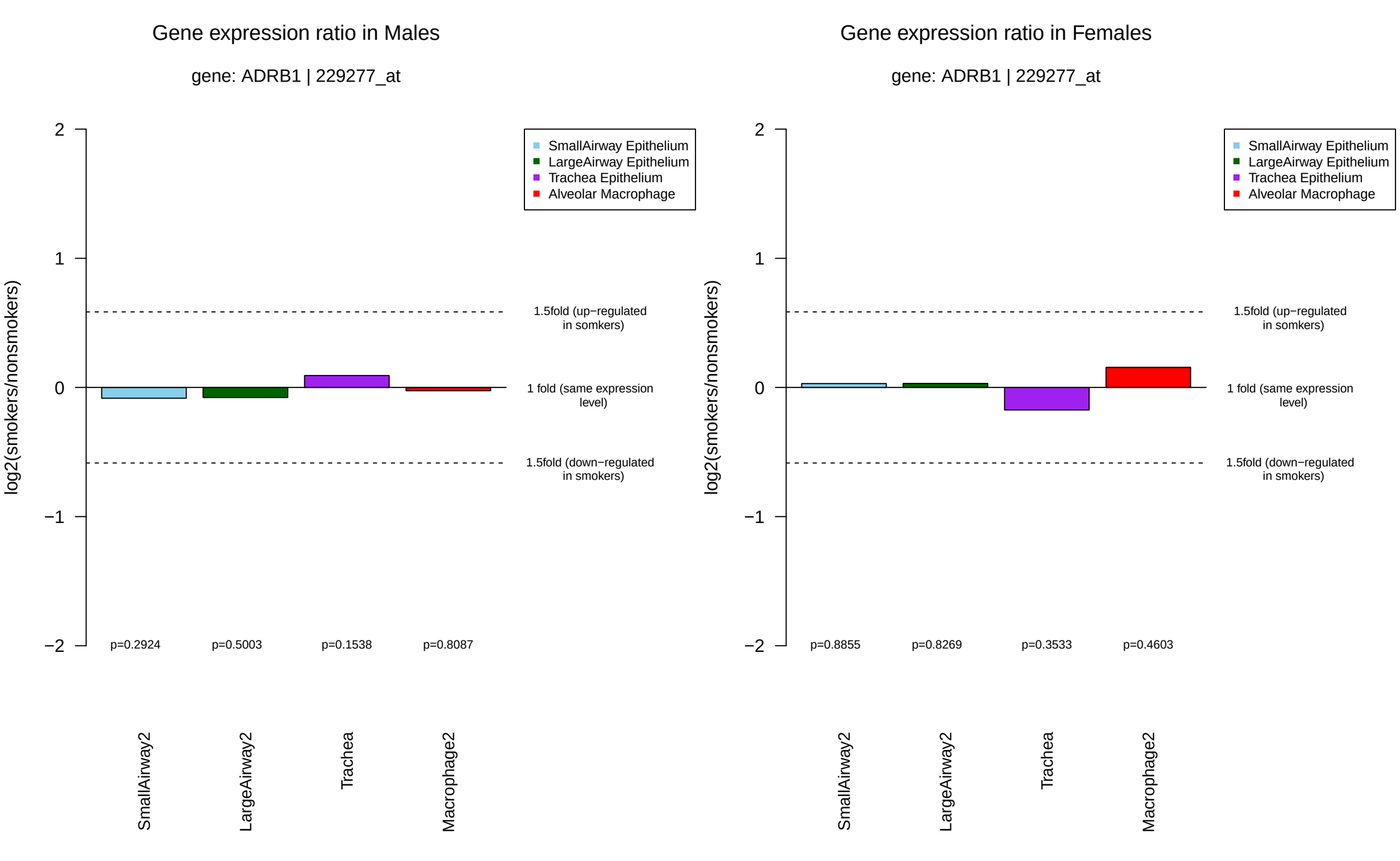
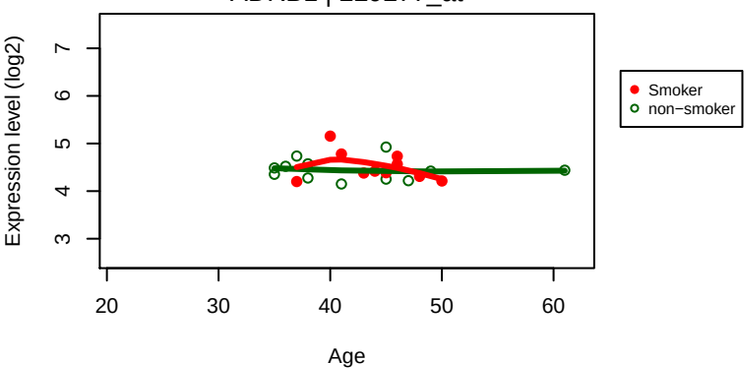
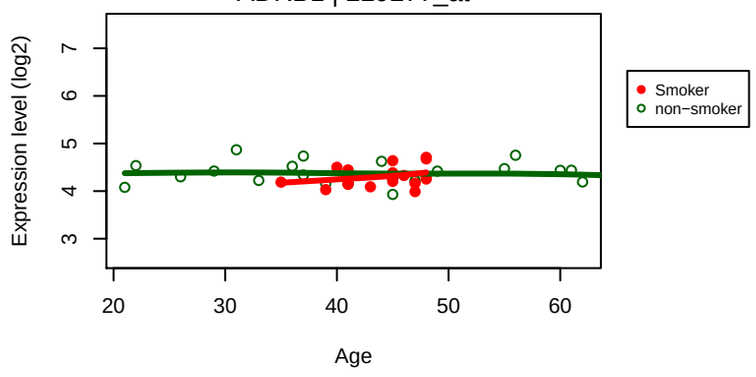
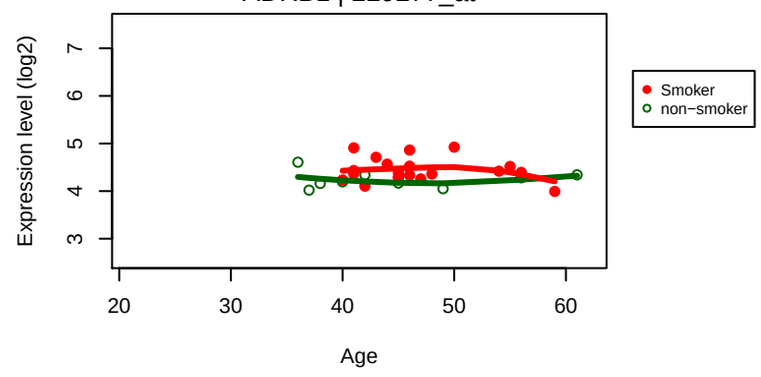

Gene expression of smokers and nonsmokers
gene: ADRB1 | 229277_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.5606
p=0.3978
p=0.9998
p=0.0216
p=0.7475
p=0.6622
p=0.8819
p=0.714
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

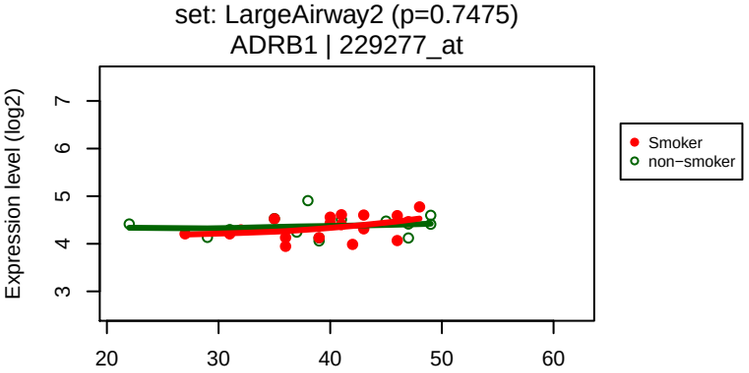

Gene expression ratio (smokers/nonsmokers)
gene: ADRB1 | 229277_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5606
p=0.3978
p=0.9998
p=0.0216
p=0.7475
p=0.6622
p=0.8819
p=0.714
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

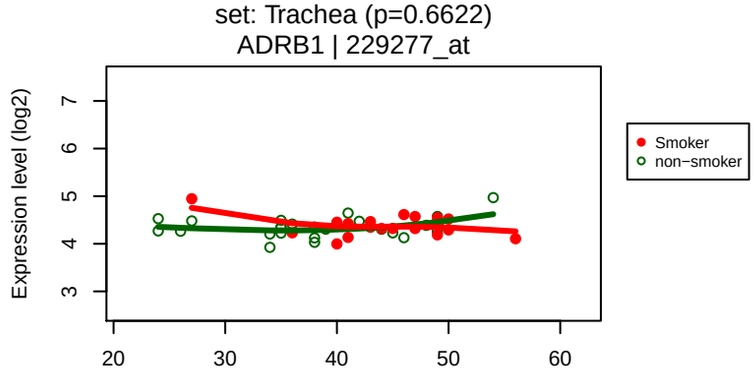

Gene expression ratio in Males
gene: ADRB1 | 229277_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2924
p=0.5003
p=0.1538
p=0.8087
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ADRB1 | 229277_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8855
p=0.8269
p=0.3533
p=0.4603
SmallAirway2
LargeAirway2
Trachea
Macrophage2

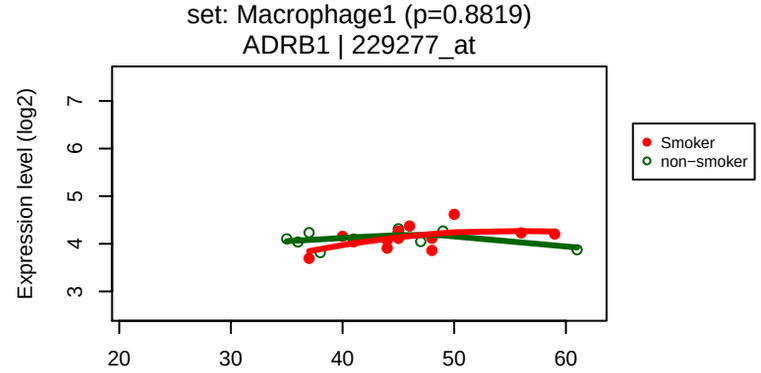

set: SmallAirway1 (p=0.5606)
ADRB1 | 229277_at
Expression level (log2)
Age
Smoker
non−smoker

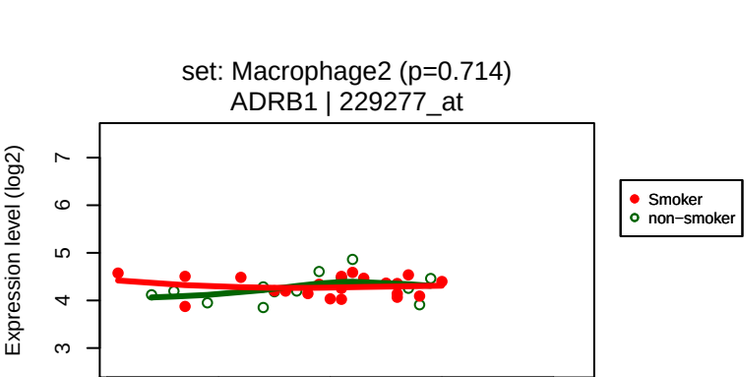

set: SmallAirway2 (p=0.3978)
ADRB1 | 229277_at
Expression level (log2)
Age
Smoker
non−smoker

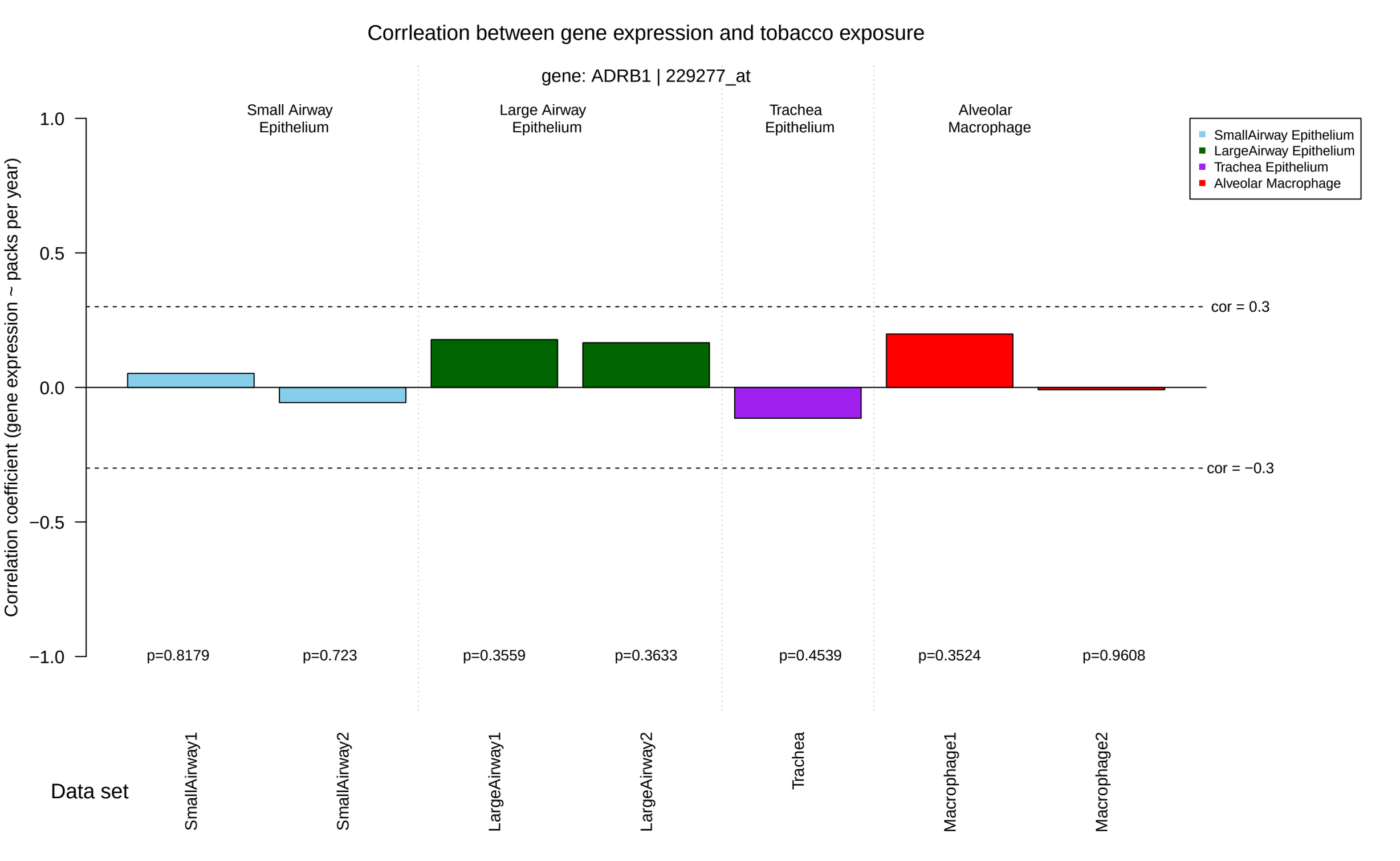

set: LargeAirway1 (p=0.0216)
ADRB1 | 229277_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7475)
ADRB1 | 229277_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.6622)
ADRB1 | 229277_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8819)
ADRB1 | 229277_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.714)
ADRB1 | 229277_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ADRB1 | 229277_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.8179
p=0.723
p=0.3559
p=0.3633
p=0.4539
p=0.3524
p=0.9608
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2